PLL MOSbius

CISL - Analog and RF IC Design Group
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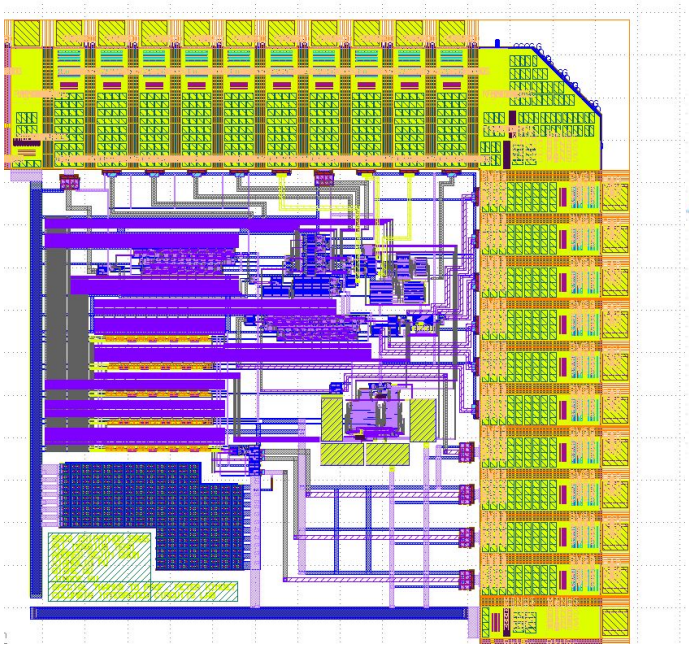
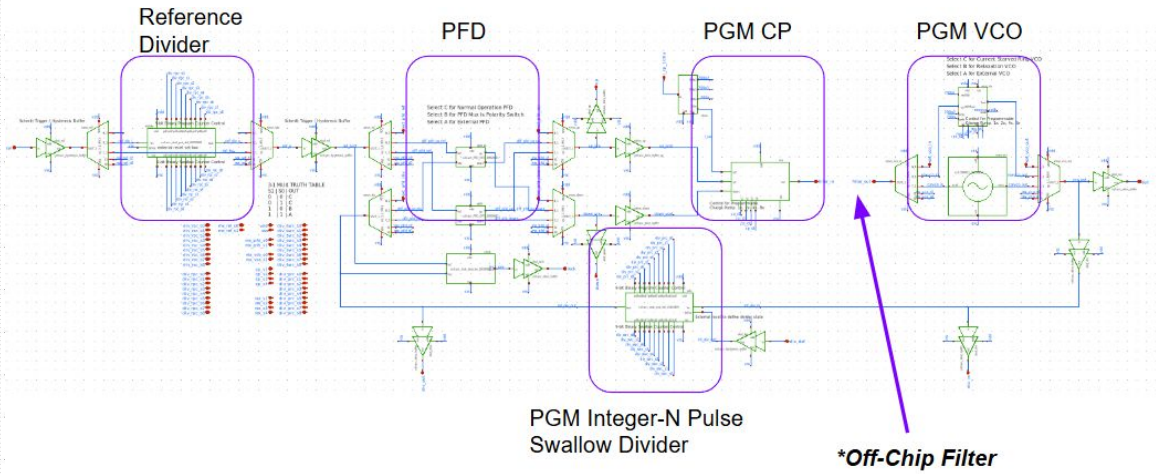
Status: Taped-Out

Abstract: Type-II, 3rd-Order Programmable Phase-Locked Loop (PLL) operable from 5 MHz to 150 MHz for AM and FM radio applications depending on the reference frequency. Includes different on-chip VCOs (current-starved ring VCO, relaxation VCO), a programmable charge pump, a programmable pulse-swallow divider, and polarity-switchable phase-frequency detector. The loop filter is implemented off-chip.





Lessons Learned and Best Practices

What worked amazingly?

- The environment (Docker) was extremely easy to download and install. It is amazing that such near complete EDA tools are freely available for all to use.
- The mentors and community members involved were all extremely helpful and supportive.
- Free tape-out experience and mentorship from experts in the IC design field is a unique opportunity that I do not think exists outside this program.

What didn't work or needs improvement

- As with all tape-outs, there was quite of bit of stress and time constraint near the end. I personally think that the layout deadlines need to be a month earlier for everyone to work out DRC/LVS issues and to start the integration process earlier.
- Final DRC checks needed to be done using commercial tools since some DRC violations were not detectable in klayout. This was a large bottleneck near the end. I cannot complain too much about free EDA tools but it would be great if the entire process could be done on the open-source environment, without the need for commercial tools.

Lessons Learned and Best Practices

What was your favorite part of the Chipathon?

- Getting the final tape-out approval from Globalfoundries.
- My favorite part will change if the chips come back and it works as intended :)

How did LLMs or tooling help (or not)?

- Did not use LLMs.

Suggestions for next year?

- Would be great if klayout DRC could be improved to include the DRC violations only caught by commercial tools like Calibre.
- Earlier deadlines for layout and integration.